ATM Preventive Maintenance Procedures

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Department: ATM Technical Operations

Overview

This document outlines standard preventive maintenance procedures for ATM machines to ensure optimal performance, minimize downtime, and extend equipment lifespan.

Daily Maintenance Checklist

Visual Inspection (5 minutes)

- Check exterior condition for damage or tampering
- Verify all panels and doors are secure
- Inspect card reader for foreign objects
- Check for error messages on screen
- Test basic transaction functionality
- Verify receipt paper availability
- Check for adequate lighting

Cash Status Check

- Review current cash levels via admin menu
- Verify all denominations available
- Check for low cash warnings
- Schedule replenishment if below 25% capacity

Weekly Maintenance Procedures

Card Reader Cleaning (10 minutes)

Required Materials:

- Approved cleaning card (Part #CC-9901)
- Lint-free cloth
- Isopropyl alcohol (99%)

Procedure:

1. Access card reader cleaning mode: Admin > Maintenance > Card Reader
2. Insert cleaning card when prompted
3. Allow card to run through 3 complete cycles
4. Inspect card reader slot for debris
5. Wipe external surfaces with alcohol and cloth
6. Test with calibration card from toolkit
7. Document cleaning in maintenance log

Touch Screen Maintenance

Cleaning Steps:

1. Power off display if possible
2. Use microfiber cloth slightly dampened with screen cleaner
3. Wipe in circular motions, no excessive pressure
4. Dry with clean microfiber cloth
5. Check for dead pixels or discoloration
6. Test touch responsiveness across all areas
7. Recalibrate if needed: Settings > Display > Calibration

Security Inspection

Anti-Skimming Check:

- Inspect card reader for foreign devices
- Check for hidden cameras near PIN pad
- Verify tamper-evident seals intact
- Test anti-skimming sensors
- Review security camera footage
- Check door sensor functionality

Monthly Maintenance Procedures

Deep Component Cleaning (45-60 minutes)

Cash Dispenser:

1. Power down ATM and open safe
2. Remove cash cassettes
3. Clean dispenser rollers with denatured alcohol
4. Inspect separation rollers for wear (replace if flat spots visible)
5. Clean optical sensors with sensor cleaning solution
6. Vacuum out dust from dispenser housing
7. Lubricate drive mechanisms per manufacturer specs
8. Reinstall cassettes and test dispense cycles

Receipt Printer:

1. Open printer access panel
2. Clean thermal print head with cleaning pen (Part #CPH-401)
3. Clean paper feed rollers with alcohol
4. Remove dust from paper path
5. Clean cutter blade assembly
6. Test print quality and alignment
7. Adjust print density if needed: Settings > Printer > Density

Cooling System:

1. Clean all cooling fan blades with compressed air
2. Clean air intake vents and filters
3. Replace cabin air filter if visibly dirty (Part #AF-1122)
4. Verify fan operation and bearing noise
5. Check internal temperature: Should be 18-28°C
6. Ensure 6-inch clearance around ATM for airflow

Network and Communication Check

Testing Procedure:

1. Run network diagnostics: Diagnostics > Network > Full Test
2. Verify ping response time < 50ms to host
3. Check for packet loss (should be < 1%)
4. Test backup communication path if available
5. Verify SSL certificate validity
6. Update security patches if available
7. Document network performance metrics

UPS Battery Test

Battery Health Check:

1. Verify UPS status indicators (green = healthy)
2. Check battery charge level: Should be 100% when on AC
3. Perform load test: Disconnect AC for 2 minutes

4. ATM should remain operational on battery
5. Reconnect AC and verify charging resumes
6. Document battery voltage and runtime
7. Replace battery if runtime < 10 minutes (Part #UPS-BAT-12V9)

Quarterly Maintenance Procedures

Component Replacement Schedule (2-3 hours)

Air Filter Replacement:

1. Locate air filter access panel (rear of ATM)
2. Remove old filter (dispose per environmental guidelines)
3. Install new filter (Part #AF-1122)
4. Ensure proper seating and airflow direction
5. Reset filter replacement counter in system

Security Seal Inspection:

1. Photograph all existing security seals
2. Check seal integrity and serial numbers
3. Replace any broken or compromised seals
4. Document new seal numbers in security log
5. Update seal tracking database

Firmware and Software Updates:

1. Backup current configuration to USB drive
2. Download latest firmware from secure portal
3. Verify firmware signature and checksum
4. Apply updates during off-peak hours
5. Test all functions post-update
6. Roll back if any issues detected
7. Document firmware version in maintenance log

Full Diagnostic Testing

Comprehensive Test Suite:

1. Run built-in diagnostic program (Hold F1+F2+F3 at boot)
2. Test cash dispenser: Empty cycle and loaded cycle
3. Test card reader: Read test cards (magnetic and chip)
4. Test receipt printer: Print test pattern
5. Test PIN pad: Verify all keys respond
6. Test touch screen: 9-point calibration test
7. Test network connectivity: Authorization test transactions
8. Test security sensors: Door, vibration, skimming detection
9. Generate diagnostic report and save to maintenance file

Annual Maintenance Procedures

Major Component Service (4-6 hours)

Dispenser Overhaul:

1. Complete disassembly of dispenser module
2. Replace separation rollers (Part #SR-7744)
3. Replace picker assembly if wear exceeds 60%
4. Replace transport belts (Part #TB-8855)
5. Clean and lubricate all moving parts
6. Replace worn solenoid valves
7. Recalibrate all sensors
8. Perform 100-cycle stress test

Receipt Printer Service:

1. Replace thermal print head if faded prints (Part #TPH-9966)
2. Replace cutter blade assembly (Part #CB-5533)
3. Replace paper feed roller assembly
4. Clean and lubricate paper path
5. Update printer firmware to latest version
6. Test with various paper types

UPS Battery Replacement:

1. Test battery capacity with load tester

2. Replace if capacity < 70% of rated (typically after 3-5 years)
3. Install new battery (Part #UPS-BAT-12V9)
4. Charge fully before testing (4-6 hours)
5. Perform runtime test under load
6. Document battery installation date

Complete Security Audit:

1. Review all access logs for past year
2. Test all security sensors and alarms
3. Verify encryption key status
4. Update security certificates
5. Review and update user access permissions
6. Test emergency shutdown procedures
7. Conduct physical security assessment
8. Update security documentation

Troubleshooting Common Maintenance Issues

Issue: Dispenser Becoming Sluggish

Symptoms: Slower cash dispensing, intermittent jams

Root Cause: Lack of lubrication, worn components

Solution:

1. Clean dispenser thoroughly
2. Apply manufacturer-approved lubricant to:
    - Drive shaft bearings
    - Picker assembly pivot points
    - Transport belt guides
3. Replace separation rollers if worn
4. Test dispense speed: Should complete in 3-5 seconds

Issue: Frequent Paper Jams

Symptoms: Paper stuck in printer multiple times per week

Root Cause: Wrong paper type, dirty rollers, high humidity

Solution:

1. Verify paper specifications: 80mm thermal, 55-65 GSM
2. Clean feed rollers with alcohol
3. Check storage conditions: 40-60% humidity, 18-25°C
4. Replace paper from different batch
5. Adjust paper tension settings if available

Issue: Touch Screen Drift

Symptoms: Touch inputs not accurate, need to press offset from target

Root Cause: Calibration drift, electrical interference

Solution:

1. Perform full touch screen calibration
2. Check for nearby EMI sources (speakers, transformers)
3. Clean touch screen thoroughly
4. Update touch screen drivers
5. Replace touch overlay if physically damaged

Maintenance Tools and Supplies

Essential Tool Kit

Tools Required:

- Multi-bit screwdriver set (Phillips, flat, Torx)
- Needle-nose pliers
- Wire cutters/strippers
- Digital multimeter
- Flashlight/inspection light
- Anti-static wrist strap
- Compressed air canister
- Vacuum cleaner (HEPA filter)

Cleaning Supplies:

- Card reader cleaning cards (Part #CC-9901)
- Isopropyl alcohol (99%)

- Denatured alcohol
- Lint-free cloths
- Microfiber cloths
- Screen cleaning solution
- Print head cleaning pen (Part #CPH-401)
- Cotton swabs
- Anti-static cleaning solution

Consumables to Stock:

- Thermal paper rolls (80mm x 80mm)
- UPS batteries (Part #UPS-BAT-12V9)
- Air filters (Part #AF-1122)
- Security seals
- Lubricant (manufacturer-approved)

Spare Parts Inventory

Critical Spares (Keep on Hand):

- Card reader module (Part #CRM-5544)
- Receipt printer assembly (Part #RPA-6677)
- Dispenser picker assembly (Part #DP-2847)
- UPS battery (Part #UPS-BAT-12V9)
- Cooling fans (Part #FAN-120MM)
- Power supply unit (Part #PSU-450W)

Regular Replacement Parts:

- Separation rollers (Part #SR-7744)
- Transport belts (Part #TB-8855)
- Print head (Part #TPH-9966)
- Cutter blade (Part #CB-5533)
- Air filters (Part #AF-1122)

Maintenance Documentation

Required Records

Maintenance Log:

- Date and time of service
- Technician name and ID
- Work performed
- Parts replaced (with serial numbers)
- Issues found and resolved
- Test results
- Next scheduled maintenance date

Transaction Monitoring:

- Daily transaction counts

- Success/failure rates
- Average transaction time
- Cash dispensed per day
- Common error codes
- Peak usage times

Parts Tracking:

- Part installation dates
- Part serial numbers
- Expected replacement dates
- Warranty information
- Supplier contact information

Safety Procedures

General Safety Rules

1. **Power Safety:**
    - Always disconnect power before internal service
    - Use UPS for emergency power during service
    - Verify zero voltage with multimeter before touching components
    - Never bypass safety interlocks
2. **Cash Handling:**
    - Follow dual-control procedures when accessing cash

- Use security cameras during all cash access
- Document all cash amounts handled
- Secure cash immediately after service

3. **Physical Safety:**
    - Wear safety glasses when using compressed air
    - Use anti-static wrist strap when handling electronics
    - Ensure proper lifting technique for heavy components
    - Keep work area clean and organized
4. **Data Security:**
    - Never leave ATM unattended during service with admin access open
    - Use secure methods to backup configuration data
    - Protect encryption keys per PCI-DSS requirements
    - Log all administrative access

Performance Metrics

Key Performance Indicators (KPIs)

Uptime Target: 99.5% or higher **Mean Time Between Failures (MTBF):** > 720 hours **Mean Time To Repair (MTTR):** < 2 hours **Preventive Maintenance Compliance:** 100%

Monthly Reporting:

- Total uptime percentage
- Number of service calls

- Types of failures
- Parts consumption
- Maintenance costs
- Customer complaints

Emergency Procedures

Critical Failure Response

If ATM Becomes Inoperable:

1. Place "Out of Service" sign immediately
2. Document failure time and symptoms
3. Check if issue is safety-related (fire, smoke, electrical hazard)
4. Contact emergency services if safety hazard present
5. Notify technical support: 1-800-ATM-TECH
6. Secure ATM if cash access compromised
7. Begin troubleshooting if safe to do so

Escalation Path:

- Level 1: On-site technician (0-30 minutes)
- Level 2: Senior technician (30-60 minutes)
- Level 3: Vendor support (1-2 hours)
- Level 4: Emergency replacement ATM (2-4 hours)

Compliance Requirements

Regulatory Standards

PCI-DSS Compliance:

- Regular security assessments
- Secure key management
- Access control logging
- Physical security measures

ADA Compliance:

- Maintain accessible height for controls
- Ensure audio guidance system functional
- Verify Braille labels present and accurate
- Test TTY/TDD compatibility

Environmental:

- Proper disposal of electronic waste
- Battery recycling per regulations
- MSDS compliance for cleaning chemicals
- Energy efficiency standards

Contact Information

Technical Support: 1-800-ATM-TECH (24/7)
Parts Ordering: 1-800-ATM-PART
Vendor Support: 1-800-VENDOR-1
Emergency Hotline: 1-800-ATM-HELP

Online Resources:

- Maintenance Portal: https://support.atm-company.com
- Parts Database: https://parts.atm-company.com
- Training Videos: https://training.atm-company.com

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